

Mathematics Reference: Logarithms

Educational Series

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1 Introduction

The logarithm of a number to a given base is the exponent to which the base must be raised to produce that number. If $a^x = n$, then $\log_a n = x$, where $a > 0$, $a \neq 1$, and $n > 0$.

2 Fundamental Laws of Logarithms

2.1 Product Rule

The logarithm of a product is the sum of the logarithms of its factors.

$$\log_a(m \times n) = \log_a m + \log_a n$$

2.2 Quotient Rule

The logarithm of a quotient is the difference between the logarithms of the dividend and the divisor.

$$\log_a\left(\frac{m}{n}\right) = \log_a m - \log_a n$$

2.3 Power Rule

The logarithm of a number raised to a power is the power multiplied by the logarithm of the number.

$$\log_a(m^n) = n \log_a m$$

3 Change of Base Formulas

To change the base of a logarithm to a new base b :

$$\log_a m = \frac{\log_b m}{\log_b a}$$

3.1 Reciprocal Property

$$\log_a b = \frac{1}{\log_b a}$$

4 Important Properties and Identities

- **Logarithm of 1:** $\log_a 1 = 0$ (since $a^0 = 1$)
- **Logarithm of Base:** $\log_a a = 1$ (since $a^1 = a$)
- **Base Power Rule:** $\log_{a^k} m = \frac{1}{k} \log_a m$
- **Identity:** $a^{\log_a n} = n$
- **Equality:** If $\log_a m = \log_a n$, then $m = n$

5 Common vs. Natural Logarithms

1. **Common Logarithm:** Base 10, usually written as $\log x$ (implied base 10).
2. **Natural Logarithm:** Base e (Euler's number ≈ 2.718), written as $\ln x$.
3. **Relationship:** $\ln x = \log_e x \approx 2.303 \log_{10} x$.

6 Solved Examples

Example 1: Simplify $2 \log 5 + \log 4$.

- Step 1: Apply Power Rule $\rightarrow \log(5^2) + \log 4 = \log 25 + \log 4$
- Step 2: Apply Product Rule $\rightarrow \log(25 \times 4) = \log 100$
- Step 3: Solve $\rightarrow \log_{10} 100 = 2$

Example 2: Solve for x in $\log_2 x = 5$.

- Convert to exponential form: $x = 2^5$
- $x = 32$